

Advanced Population Modeling : Integrated Analysis / Integrated Population Modeling

(Newman et al. Chapter 9, Zipkin & Saunders 2018, Maunder & Punt 2013) Acknowledgements: Steve Cadrin

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Outline

This week

- ▶ Questions on HW3, Jason's presentations
- ▶ Questions from Reading
- ▶ Integrated Population Modeling / Analysis
- ▶ Revisit Age-structured modeling in RTMB

Integrated Population Modeling

- ▶ We have been building toward this
- ▶ Principle throughout class, fitting models to multiple data sources simultaneously
- ▶ As opposed to doing separate analyses for different types of data, parameter types
- ▶ Often shared latent variables (the population process), treating together makes our modeling approach more self-consistent and allows for integrated variance assumptions/calculations.

Outline

- Statistical Catch at Age
 - Population model component
 - Observation model component
 - Statistical model component
 - Integrated Analysis
 - More complex population model component
 - Expanded observation model component
 - Weighted statistical model components
 - Internal calculation of reference points
 - Stock Synthesis Example
- 



More Complex Population Models

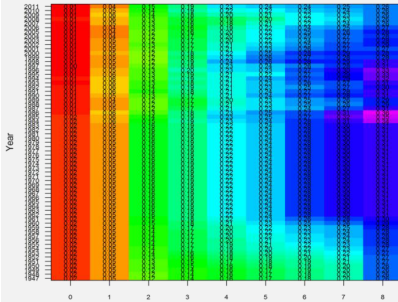
- Size structure
 - direct use of most available information
 - complements samples of age at size
 - Selectivity is usually size-based
 - Population model includes age structure and size distribution at age
- Spatial structure
 - Accounts for resource distribution and fishery patterns
 - Can account for movement among areas
- Seasonal structure
 - Accounts for seasonality of fishing
 - Accounts for growth patterns and time of samples
- Sex structure
 - Allows for different life histories (growth, natural mortality, ...)



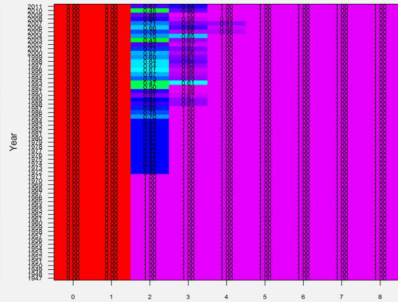
Empirical Approach to Growth, Maturity, Fecundity

- Requires representative sampling each year.

Empirical weight at age in middle of the year

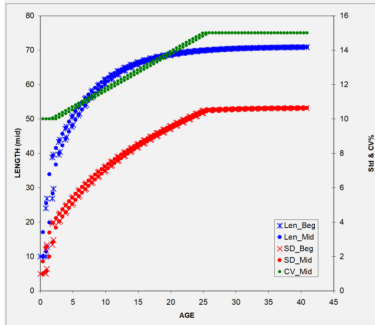


Maturity x fecundity

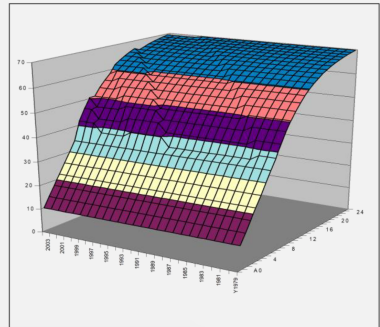


Complex Growth Models

- Two distinct recruitment events per year



Temporal variation



Extensive Observation Models

- Model prediction of catch at length, sex, geographic zone, and seasonal period:

$$\hat{C}_{t,l,s,z,p} = \frac{N_{t,l,s,z,p} F_{t,l,s,z,p} (1 - e^{-(M_{t,l,s,z,p} + F_{t,l,s,z,p})})}{(M + F_{t,l,s,z,p})} + \varepsilon_{t,l,s,z,p}$$

- Model prediction of tag recoveries (r) as a function of tagged fish at large (n) and reporting rate (β):

$$\hat{r}_{t,l,s,z,p} = \frac{n_{t,l,s,z,p} F_{t,l,s,z,p} (1 - e^{-(M_{t,l,s,z,p} + F_{t,l,s,z,p})})}{(M + F_{t,l,s,z,p})} \beta + \varepsilon_{t,l,s,z,p}$$

Dis-Integrated Analysis

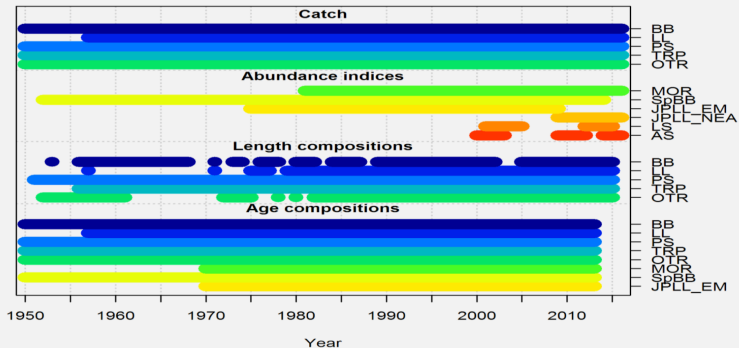
- Pre-Processed Input 'data'
 - Aggregate catch-at-age – all fleets, discards plus landings, expanded age samples to length samples then to total landings and discards, often 'patching' gaps with missing data
 - Standardized CPUE – GLM of sample data
- Estimation Models
 - Biomass dynamics model – longer time series of catch and CPUE or surveys
 - VPA – shorter series of catch-at-age
 - Analysis of tagging data – estimation of mortality or movement
- Post-Processed Reference Point Models – some of the same inputs (M, weight at age, maturity) and results from the estimation model (selectivity, SSB, R) are used to determine management limits or targets that are consistent with estimates.

Integrated Model

- Attempts to model sample observations directly:
 - E.g., monthly length composition by fleet, monthly age-at-length composition
 - Missing data is not a problem.
 - Fit to data when it is available (early period with only catch, later periods with more indices and composition data)
- Simultaneous analysis provides consistent (and comparable) parameter values and assumptions.
- Integrated analysis of all data with shared parameter estimates allows for more comprehensive measures of uncertainty.

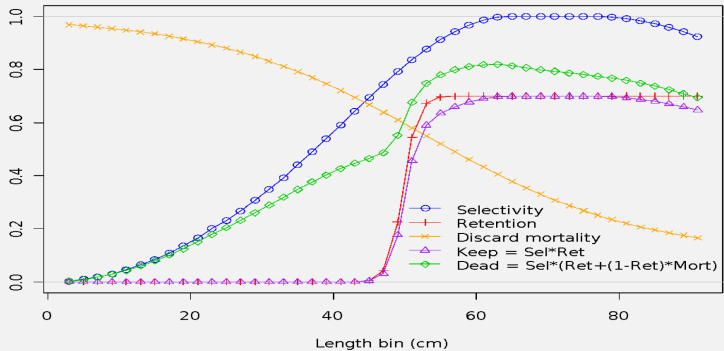
Heterogeneous Data with Gaps

Data by type and year



Integrated Approach to Estimating Discards

Male ending year selectivity for fishery1



Model Expectations of Tagging Data

- Predicted fishery recoveries (r) are a function of the number of tags at large (n), the fishery's exploitation rate, and reporting rate (β), and the assumed error distribution is typically a multinomial frequency.

$$r_t = n_t \frac{F_t}{F_t + M} [1 - e^{-(F_t + M)}] \beta$$



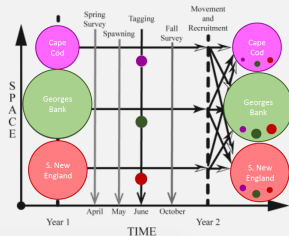
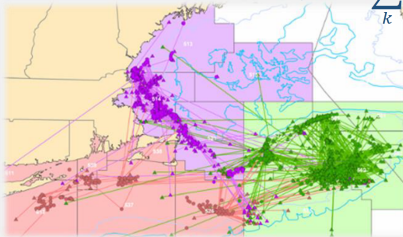
Release year	Number of releases	1	2	3	4
2003	4392	317	115	26	10
2004	2811	0	79	52	13
2005	1121	0	0	59	13
2006	3287	0	0	0	253

Multiple Areas



- Three adjacent stocks (k) of yellowtail flounder (Goethel et al. 2015).
 - Untagged and tagged fish can reside in the same area (j) or move to adjacent areas (k) each year, with estimates of annual transfer rates among areas (T).

$$N_{t+1,a+1} = \sum_k T_{j \rightarrow k} N_{t,a} e^{-(F_{t,a} + M)}$$



Statistical Model

- Some SCAA models predict $C_{t,a}$ assuming lognormal error (with least squares of likelihood)
- Most SCAA apply Fournier & Archibald's (1982) approach of predicting C_t with lognormal likelihood and $P_{t,a}$ with multinomial likelihood

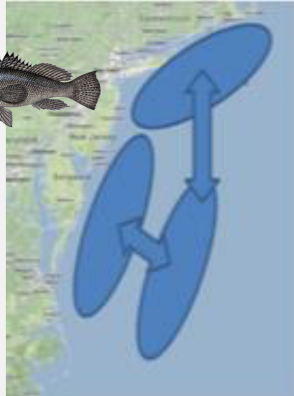
$$-Ln(L) = nLn(\sigma) + \sum \frac{[Ln(C_t) - Ln(\widehat{C}_t)]^2}{2\sigma^2}$$

$$-Ln(L) = ESS \sum_{t,a} P_{t,a} Ln(\widehat{P}_{t,a})$$

- *ESS*: effective sample size of age composition data
- ... in addition to other $-Ln(L)$ components (e.g., predicted index, tag recoveries, ...)

Black Sea Bass

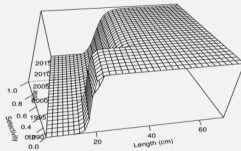
- The 2012 statistical catch at age model was rejected, because it ignored spatial structure and hermaphroditism.
- Fay et al. (2016) developed a stock synthesis application to sea bass with sex, geographic regions, seasons, size,



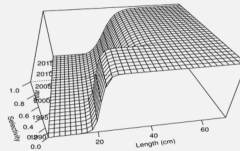
Selectivity



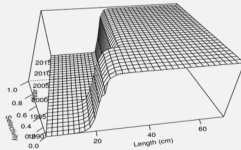
Female time-varying selectivity for North_Trawl_1



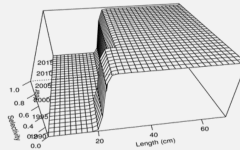
Female time-varying selectivity for North_Trawl_2



Female time-varying selectivity for North_Nontrawl_1



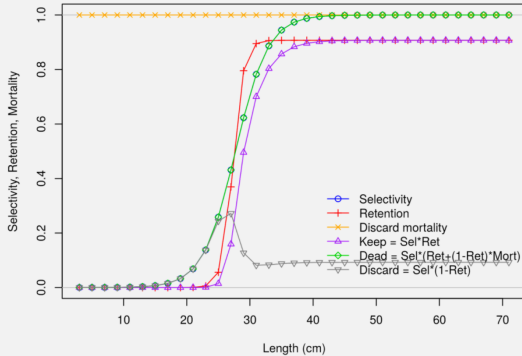
Female time-varying selectivity for North_Nontrawl_2



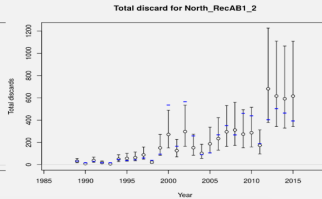
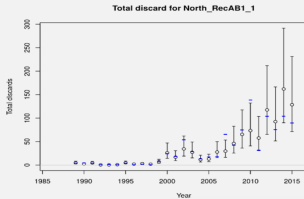
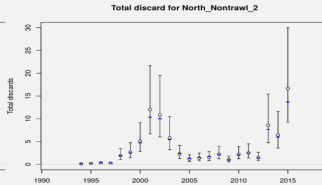
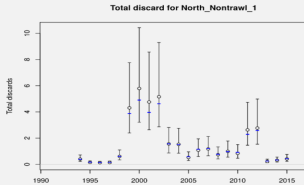
Size Retention



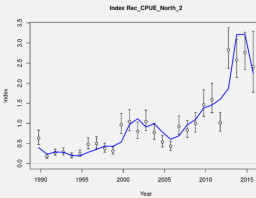
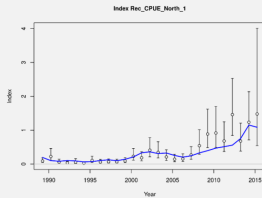
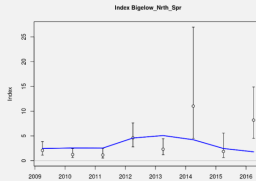
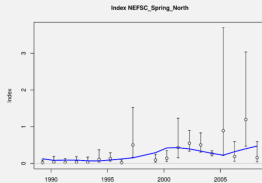
Female ending year selectivity for North_Trawl_1



Observed and Predicted Discards

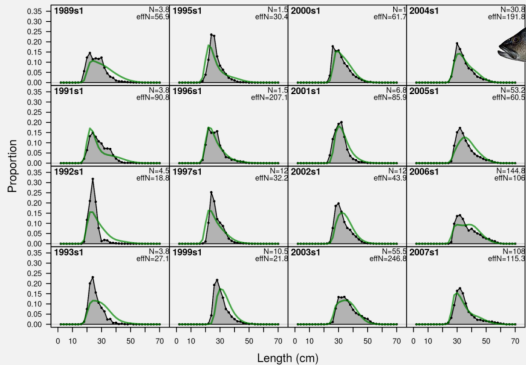


Observed and Predicted Stock Size Indices



Observed and Predicted Length Composition

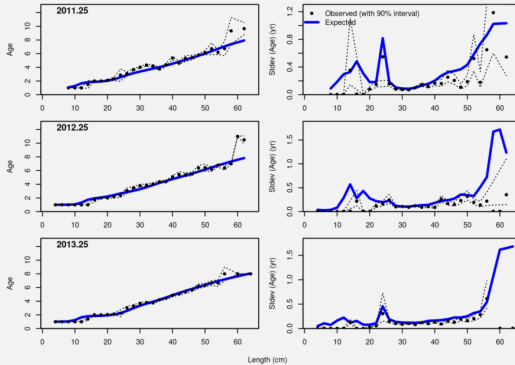
length comps, retained, North_Trawl_1



Observed and Predicted Age at Length



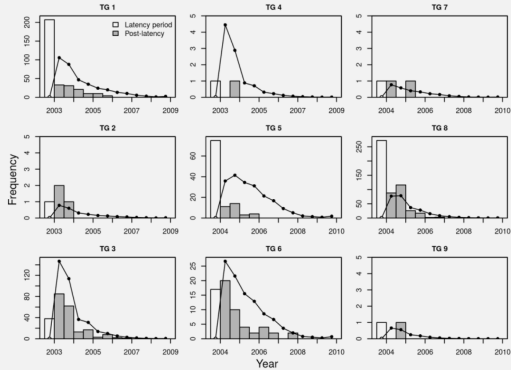
Conditional AAL plot, whole catch, South_trawl_1



Model Fit – Tag Recoveries



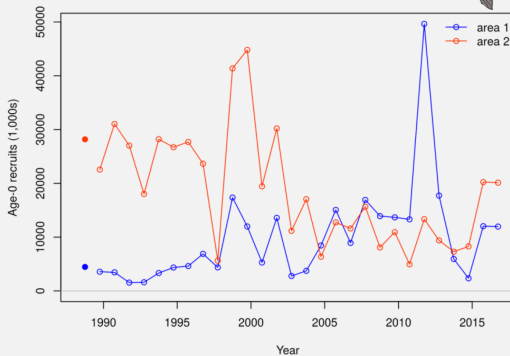
Fit to tag recaptures by tag group



Regional Recruitment



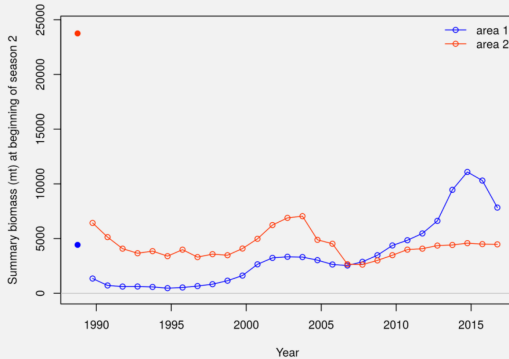
Age-0 recruits (1,000s) by area



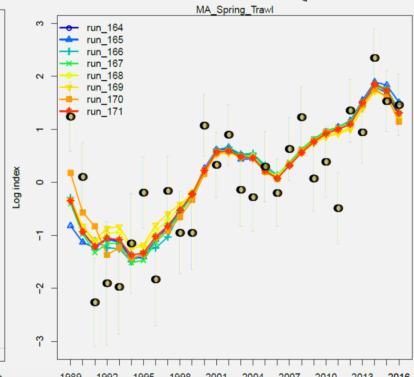
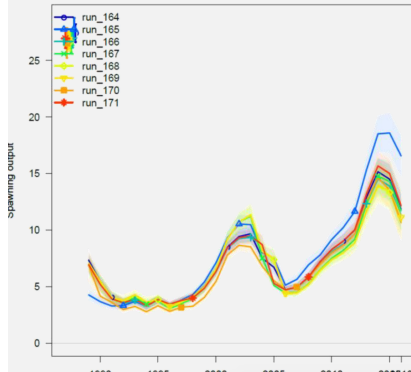
Regional Biomass



Summary biomass (mt) at beginning of season 2 by area



Sensitivity Analysis



Integrated Analysis

- ▶ Often have complicated population model components, but can fit to a range of data conditions, from data-limited to data-rich
 - ▶ Can fit to disaggregated directly
 - ▶ May be able to match system mechanisms and data generating processes more closely (e.g. model length-based processes as such, ie selectivity, retention, mortality, etc.
- ▶ Can allow for direct inclusion of range of population drivers, environmental factors, etc.
- ▶ Statistical estimation is more challenging
 - ▶ thousands(!) of estimated parameters
 - ▶ many likelihood components

“If applying a VPA is like driving a car, then developing an integrated model is like flying a helicopter”